

Chapter 1

AIR FORCE TERMINAL AERODROME FORECAST (TAF) CODE

1.1. Roles and Responsibilities. The procedures in this manual apply to all AF Weather organizations or associated contractors performing forecasting of weather and solar operations in support of AF, Army or DoD wide operations. Compliance items for this publication are driven by international policy to ensure safety of aviation. This chapter gives instructions for encoding Terminal Aerodrome Forecasts (TAFs). Air Force weather organizations specify, amend, and disseminate TAFs in accordance with AFI 15-128, *Weather Roles and Responsibilities* and AFMAN 15-129 Volume 1, *Air and Space Weather Operations – Characterization*.

1.2. Code Format. Air Force weather forecast coding practices are derived from international standards established by the World Meteorological Organization (WMO) as published in WMO No. 306, *Manual on Codes*, Volume I.1, Part A, Section FM 51, Aerodrome Forecast, and aligns with practices of the Aviation Routine Weather Report (METAR) code found in AFMAN 15-111, *Surface Weather Observations*.

1.2.1. Unless otherwise specified, forecast elements in the main body of the forecast text (clouds, weather, wind, etc.) apply to the area at or within a 5 statute mile radius of the center of the aerodrome. Specified weather greater than 5 statute miles but less than or equal to 10 statute miles of the aerodrome center is encoded as VC (in the vicinity). **(T-0)** Do not specify elements outside of the “vicinity” in forecasts.

1.2.2. Forecast elements represent the expected condition during the forecast period and in the forecast area.

1.3. TAF Encoding.

1.3.1. TAF Code Format. Use the following format in [Figure 1.1](#) for encoding TAFs:

Figure 1.1. TAF Code Format.

MESSAGE HEADING
TAF (AMD or COR) CCCC YYGGggZ YYG1G1/YYG2G2 dddffGfmfmKT VVVV
w’w’ NsNsNshshsCC or VVhshshs or SKC (VAbbbttt) (WShxhxhx/dddfffKT)
(6IchihihitL) (5BhBhBhBtL) QNHP1P1P1P1INS (Remarks)
TTTTT YYGGGGeGe or YYGG/YYGeGe dddffGfmfmKT...same as above... (Remarks)
TX(M)TFTE/YYGFGFZ TN(M)TFTE/YYGFGFZ

1.3.1.1. Make all TAFs valid for a 30-hour forecast period.

1.3.1.2. Use groups in parentheses only as condition exists or as required.

1.3.2. Specification of Symbolic Letters.

1.3.2.1. Message Heading (TAF [AMD or COR] CCCC YYGGggZ YYG1G1/YYG2G2). The message heading consists of:

1.3.2.1.1. Message identifier of TAF

1.3.2.1.2. Forecast modifier indicating an amendment or correction (AMD or COR). Only one modifier at a time.

1.3.2.1.2.1. When issuing an amendment (AMD), issue only the remaining valid period of the TAF (e.g. If a TAF originally starting at 1600Z is amended at 1847Z, only forecast groups valid at and after 1800Z are included; groups that are no longer valid are removed).

1.3.2.1.2.2. When issuing a correction (COR), issue the entire original text of the TAF, changing only the TAF header and the erroneous elements (e.g., if a TAF originally starting at 1600Z is corrected at 1615Z, all forecast groups remain included). See [Figure 1.4](#) for an example TAF correction.

1.3.2.1.3. Four Letter Location identifier (CCCC)

1.3.2.1.4. Issue Date and Time, YYGGggZ. The issue date and time consists of the current day of the month (YY) and the Coordinated Universal Time (UTC) in hours (GG) and minutes (gg) followed by the letter Z. This time is updated for each change to a TAF (i.e. a new TAF, amendments, and/or corrections).

1.3.2.1.5. Valid Period (YYG1G1/YYG2G2). The valid period consists of the current day of the month (YY) and the 30-hour period of the forecast beginning time (G1G1) and ending time (G2G2) in whole hours, except for amended TAFs. All times are in UTC. For TAF groups starting and stopping at midnight UTC, use 00 and 24, respectively, to indicate the appropriate valid times. Amended TAFs are valid from the current hour to the ending hour of the original TAF. For example, if the current time is 1640Z, the amended time would be 16Z; if the current time is 2110Z, the amended time would be 21Z. For example, amending the 0318/0424Z TAF at 2131Z, the valid period is 0321/0424Z.

1.3.3. Change Groups (TTTTT). Use BECMG YYGG/YYGeGe, TEMPO YYGG/YYGeGe, and FM YYGGgg change groups to indicate changes from the predominant forecast condition at some intermediate date and hour time (YYGGgg) or during a specified period between hours (YYGG to YYGeGe). TEMPO groups may be used to forecast a change in any or all forecast groups and be followed by a description of all the elements for which a change is forecast to occur intermittently from YYGG to YYGeGe. **Exception:** Non-convective low-level wind shear and QNH groups are not included in TEMPO groups. FM change groups must include all encoded elements. Start a new line of text for each change group. Change groups that begin or end at midnight UTC will use 00 and 24 respectively to indicate the appropriate valid times. Limit change groups to those that are significant to airfield operations. **(T-0)** Avoid overlapping forecast periods in order to avoid confusion and keep the intent of the forecast simple.

1.3.3.1. Becoming (BECMG)—The change-indicator group TTTTT YYGG/YYGeGe in the form of BECMG YYGG/YYGeGe is used to indicate a change to forecast prevailing conditions expected to occur at either a regular or irregular rate at an unspecified time within the period defined by a two-digit date (YY), two-digit change beginning time (GG) with a slash separating a two-digit date (YY) and a two-digit ending time (GeGe) in whole hours. The time-period described by a BECMG group is usually for one hour but never exceeds two hours. This change to the predominant conditions are followed by a description of all elements for which the change is forecast. The forecast conditions encoded after the BECMG YYGG/YYGeGe group are those elements expected to prevail from the ending time of this change group (GeGe) to the ending time of the forecast period

(YYG2G2). The forecasted conditions must occur less than 30 minutes after the YYGeGe group start time. **(T-0)** When using the BECMG group to forecast a change in one or more elements, repeat the entire element(s). For example, if the BECMG group was used to forecast a decrease in the ceiling and all other forecast layers were expected to remain the same, the entire cloud code group is repeated, not just the ceiling layer.

1.3.3.2. Temporary (TEMPO)—The change-indicator group TTTTTT YYGG/YYGeGe in the form of TEMPO YYGG/YYGeGe group is used to indicate temporary fluctuations to the forecast meteorological conditions. Conditions described by the TEMPO group must occur once during the specified time-period indicated by the date YY and time GG to the date YY and time G_eG_e, for less than 30 consecutive minutes or occur for an aggregate total of less than 30 minutes of every cardinal hour and cover less than half of the period indicated by the date YY and time GG to the date YY and time G_eG_e. **(T-1) Exception:** Organizations will allow 45 minutes for thunderstorms. **(T-0)** The extra 15 minutes provide for the 15-minute period between the time thunder is last heard and the time the thunderstorm is officially ended. If forecast conditions in the TEMPO group last more than 30 consecutive minutes or are expected to last more than half of the period indicated by the time YYGG to YYGeGe, then the temporary condition will be considered to be predominant and entered in the initial forecast period or following a FMYYGGgg or BECMG group. **(T-0)**

1.3.3.3. From (FMYYGGgg)—The time indicator YYGGgg in the form of FMYYGGgg is used to indicate the beginning of a self-contained part of the forecast indicated by the two-digit date YY and four-digit time GGgg. When the group FMYYGGgg is used, all forecast conditions preceding this group are superseded by the conditions forecasted in this group. For example, if the TAF period is 1909/2015 and a change is forecast at 1420 UTC, the entry FM191420 shall be encoded. The elements entered on this line are in effect from 191420 UTC to the end of the forecast period, 201500 UTC. While the use of a four-digit time in whole hours (e.g., 1600) remains acceptable, a forecast and amending events may require a higher time resolution. Use forecast minutes in this case. Four-digit resolution will only be used in this FMGGgg group. The forecasted conditions must occur in less than 30 minutes from the time specified in the YYGGgg group. **(T-0)**

1.3.4. Wind Group (dddffGfmfmKT). Surface wind direction, speed and gusts, if any.

1.3.4.1. Wind direction (ddd). Forecast true wind direction (from which wind is blowing) to the nearest 10 degrees. If direction varies more than 60 degrees, encode the prevailing direction for ddd and append the limits of variability to remarks (e.g., WND 270V350). Forecast a prevailing wind direction whenever it can be determined. In rare cases, there may be situations when forecasting a prevailing direction is not possible. In these situations, encode VRB for ddd.

1.3.4.1.1. When winds are calm, encode dddff as 00000KT.

1.3.4.1.2. When wind speed is 6 knots or less and a direction cannot be determined, encode dddff as VRBff.

1.3.4.2. When wind speeds are be more than 6 knots, do not use VRB for ddd unless the situation involves air-mass thunderstorm activity during which forecasting a prevailing wind direction with confidence is not possible. When it is possible to forecast the peak gust direction, but not the prevailing direction, encode the wind group as VRBffGfmfmKT and append the probable peak gust direction to remarks (e.g., GST DRCTN 250).

1.3.4.2.1. Wind Speed (ff). Mean forecast wind speed in whole knots. When speed is equal to or greater than 100 knots, use three digits.

1.3.4.2.2. Gusts (Gfmfm). Forecast speed or gusts, in whole knots. Encode gusts when the maximum speed exceeds a mean speed (ff) by 10 knots or more or when the peak wind speed is forecast to exceed the lull by 10 knots or more. Encode gusts of 100 knots or more in three digits. **(T-0)**

1.3.4.2.3. KT. Unit indicator for wind speeds in knots.

1.3.5. Visibility Group (VVVV). Forecast prevailing visibility in meters, rounded down to the nearest reportable value from [Table 1.1](#). Include weather and/or an obscuration (w'w') whenever visibility is forecast less than 9,999 meters. **(T-0)** If visibility alternates frequently from one significant value to another, describe the situation with a TEMPO group; do not use variable visibility remarks. **Note:** While a visibility of less than 9,999 meters requires a weather and/or obscuration, weather such as precipitation does not require a restriction to visibility to be reported in a forecast (e.g., 9999 –RA). In this case, the weather is significant because it is occurring, not because it is restricting visibility.

Table 1.1. Visibility (VVVV).

Statute Miles	Meters	Statute Miles	Meters
0	0000	1 3/8	2,200
1/16	0100	1 1/2	2,400
1/8	0200	1 5/8	2,600
3/16	0300	1 3/4	2,800
1/4	0400	1 7/8	3,000
5/16	0500	2	3,200
3/8	0600	-	3,400
-	0700	2 1/4	3,600
1/2	0800	-	3,700
-	0900	2 1/2	4,000
5/8	1,000	2 3/4	4,400
-	1,100	-	4,500
3/4	1,200	-	4,700
-	1,300	3	4,800* See NOTE 1
7/8	1,400	-	5,000* See NOTE 1
-	1,500	4	6,000
1	1,600	-	7,000
-	1,700	5	8,000
1 1/8	1,800	6	9,000
1 1/4	2,000	7 and above	9,999
Note 1: Substitute 5000 meters for 4800 meters Outside the Continental United States (OCONUS) locations based on the host-nation national practice.			

1.3.6. Forecast Weather and Obscuration Group (w'w'). AFMAN 15-111 defines forecast weather and obscurations for construction of w'w' groups ([Table 1.2](#)).

Table 1.2. Weather (w'w') Group Code.

QUALIFIER		WEATHER PHENOMENA		
INTENSITY OR PROXIMITY	DESCRIPTOR	PRECIPITATION	OBSCURATION	OTHER
1	2	3	4	5
- Light	MI Shallow	DZ Drizzle	BR Mist	PO Well-developed Dust/Sand Whirls
Moderate	PR Partial (covering part of the aerodrome)	RA Rain	FG Fog	
+ Heavy (well-developed in the case of tornadoes or waterspouts)	BC Patches	SN Snow	FU Smoke	SQ Squalls
VC In the Vicinity	DR Low Drifting	SG Snow Grains	VA Volcanic Ash	FC Funnel cloud(s) (Tornado or Waterspout)
	BL Blowing	IC Ice Crystals (Diamond Dust)	DU Widespread Dust	
	SH Shower(s)	PL Ice Pellets	SA Sand	SS Sand storm
	TS Thunderstorm	GR Hail	HZ Haze	DS Dust storm
	FZ Freezing (Super-cooled)	GS Snow Pellets	PY Spray	

1.3.6.1. Construct predominant forecast weather (w'w') groups by considering [Table 1.2](#), columns one to five in sequence. That is intensity/proximity, followed by description, followed by precipitation type (two precipitation types can be used in the same w'w' group), obscuration, or other weather phenomena (e.g., +SHRA is heavy showers of rain, +TSRAGR is thunderstorms, heavy rain, and hail; -RASN is light rain and snow; TS is thunderstorm without precipitation).

1.3.6.2. Only one w'w' group is normally included unless one group does not adequately describe the forecast situation. When more than one weather or obscuration condition exists, limit the w'w' group to three groups. When more than three w'w' groups apply to a situation, select and encode the three w'w' that are most significant to operations.

1.3.6.3. When applicable, funnel clouds (FC) and tornadoes (+FC) take precedence over all other w'w' groups and are forecast as at the station and not in the vicinity.

1.3.6.4. VC may be encoded in combination with thunderstorms (TS), showers (SH), fog (FG), blowing snow (BLSN), blowing dust (BLDU), blowing sand (BSA), well-developed dust/sand whirls (PO), sand storm (SS), and dust storm (DS). When encoding, place VC before the precipitation, obscuration, or other weather phenomena entry without

a space between the two (e.g., VCSH, VCPO). Do not encode intensity qualifiers with VC. Forecast weather in the vicinity is the last entry in the weather (w'w') group.

1.3.6.5. When an encoded predominant forecast condition is followed by a change group (BECMG or FM) without a w'w' group, encode the change group w'w' as NSW (no significant weather) to indicate that significant weather is no longer expected. This includes weather forecast in the vicinity (e.g., VCSH was included in a previous group, and forecasted to end).

1.3.6.6. Forecast Volcanic Ash (VA) as present weather regardless of restrictions to visibility when VA is observed **(T-0)** or the Volcanic Ash Advisory Center (VAAC) forecast includes a surfaced-based VA plume. **(T-1)**

1.3.6.7. Forecast Squall (SQ) when a strong wind characterized by a sudden onset in which the wind speed increases at least 16 knots and sustained at 22 knots or more for at least one minute.

1.3.7. Cloud and Obscuration Group (NsNsNshshsCC). Report as often as necessary to indicate all forecast cloud layers up to the first overcast layer. Arrange groups in ascending order of cloud bases AGL (e.g., lowest base first). **(T-0)**

1.3.7.1. Cloud Amount (NsNsNs). The cloud amount is given as sky clear (SKC = no clouds); few (FEW = trace to 2/8ths); scattered (SCT = 3/8ths to 4/8ths); broken (BKN = 5/8ths to 7/8ths); or overcast (OVC = 8/8ths). Follow the three-letter abbreviations with the height of the base of the cloud layer (mass) hshshs without a space (e.g., FEW100, SCT250). The summation principle applies. This principle states that the sky cover at any level is equal to the summation of the sky cover of the lowest layer, plus the additional sky cover at all successively higher layers, up to and including, the layer being considered. Do not assign a sky cover to a layer less than a lower layer (e.g., SCT015 FEW020 should be SCT015 BKN020).

1.3.7.2. When the sky is totally obscured, encode VVhshshs, where VV is the indicator and hshshs is the vertical visibility in hundreds of feet (e.g. VV002).

1.3.7.3. Ceiling Height (hshshs). A ceiling is the height above the earth's surface of the lowest layer reported as broken or overcast; or the vertical visibility into an indefinite ceiling. Consider all layers and obscuring phenomena to be opaque.

1.3.7.4. Indefinite Ceiling (VVhshshs). The vertical visibility measured in feet, into a surface-based total obscuration, which hides the entire celestial dome (8/8ths).

1.3.7.5. Surface-Based Partial Obscuration. When forecasting a surface-based partial obscuration, encode as FEW000, SCT000, or BKN000 as appropriate to indicate a surface-based partial obscuration. Code as a remark the obscuring phenomena and the applicable layer. For example, FG SCT000 would indicate the w'w' weather element causing the obscuration is caused by fog and layer amount is SCT. Include the amount of partial obscuration in your sky cover summation computation. Do not consider surface-based partial obscurations as a ceiling.

1.3.7.6. Variable Sky Condition. If two or more significant sky conditions alternate frequently from one to the other, describe the situation with a TEMPO group; do not use variable sky condition remarks.

1.3.7.7. Height of Cloud Base (hshshs). Forecast the height of the base of each sky cover layer in hundreds of feet AGL using the reportable layers defined in [Table 1.3](#).

Table 1.3. Reportable Cloud Layers.

Range of Height Values (feet)	Reportable Increments (feet)
< 50 feet	Round down to 000 feet
> 50 feet but ≤ 5,000 feet	To the nearest 100 feet
> 5,000 feet but ≤ 10,000 feet	To the nearest 500 feet
> 10,000 feet	To the nearest 1,000 feet

1.3.7.8. Cloud Type (CC). The only cloud type included in the aerodrome forecast is cumulonimbus (CB); when appropriate, the contraction CB follows cloud or obscuration height (hshsh) without a space. **(T-0)**. The cloud or obscuration group will include a forecast cloud type of cumulonimbus (CB) whenever a thunderstorm is included in the significant weather group. This includes forecasts for thunderstorms in the vicinity (i.e., VCTS). **(T-0)** The following example shows the use of the CB contraction:

Figure 1.2. TAF Example Using CB Contraction.

**TAF CCCC 101555Z 1016/1122 24025G35KT 0800 TSRA BKN035CB OVC080 QNH2978INS
BECMG 1017/1018 27010G15KT 9999 VCTS FEW040CB SCT080 QNH2989INS
BECMG 1019/1020 31012KT 9999 NSW SCT080 QNH2995INS TX14/1022Z
TN09/1113Z**

1.3.8. Example TAF. [Figure 1.3](#) provides an example TAF for Barksdale AFB, LA. Interpretation and explanations of the coded information follows.

Figure 1.3. TAF Example.

**TAF KBAD 011555Z 0116/0222 03008KT 0800 PRFG FEW000 BKN005 BKN012 QNH3001INS FG FEW000
TEMPO 0118/0121 14012G18KT 3200 -SHSN BLSN FEW000 OVC006 620065 BLSN FEW000
FM012145 15012G20KT 9999 NSW OVC030 QNH2992INS
BECMG 0123/0124 15012G20KT 3200 -SN BLSN FEW000 OVC004 620046 QNH2983INS BLSN FEW000
TEMPO 0201/0203 13015G25KT 0200 -FZDZ FG VV001 660001 650109 TX00/0121Z
TNM01/0212Z**

1.3.8.1. The forecast is for Barksdale AFB, LA (KBAD), issued on the first at 1555Z, valid from 011600Z to 022200Z. The initial conditions (1600Z to 2144Z) are for winds from 030 degrees at 8 knots, visibility 800 meters in partial fog; sky cover is few (either a surface-based partial obscuration or a layer at or lower than 50 feet), sky is broken (ceiling) at 500 feet and broken at 1200 feet. The lowest altimeter setting between 011600Z and 012144Z is 30.01 inches of mercury. There is a fog-induced surface-based partial obscuration from 1/8 to 2/8 coverage.

1.3.8.1.1. Between 011800Z and 012100Z, conditions vary temporarily to winds from 140 degrees at 12 knots gusting to 18 knots, visibility 3200 meters in light snow showers and blowing snow, sky cover is few at the surface (surface-based partial obscuration), overcast at 600 feet (the ceiling), with light rime icing from 600 to 5600 feet above ground level (AGL). The surface-based partial obscuration (FEW000) is caused by blowing snow.

1.3.8.1.2. Beginning at 012145Z, conditions change to wind from 150 degrees at 12 knots gusting to 20 knots; unrestricted visibility 9,999 meters or greater, no significant weather, sky cover overcast at 3000 feet and the lowest altimeter setting from 012145Z until 012400Z is 29.92 inches of mercury.

1.3.8.1.3. Between 012300Z and 012400Z, conditions become: wind from 150 degrees at 12 knots gusting to 20 knots; visibility 3200 meters in light snow and blowing snow, sky cover is few (either a surface-based partial obscuration or a layer at or lower than 50 feet), sky has an overcast ceiling at 400 feet. There is light rime icing from 400 to 6400 feet AGL and the lowest altimeter setting from 020000Z until 022200Z is 29.83 inches of mercury. There is a blowing snow-induced surface based partial obscuration from 1/8 to 2/8 in coverage.

1.3.8.1.4. Between 020100Z and 020300Z, conditions vary temporarily to winds from 130 degrees at 15 knots gusting to 25 knots, visibility 200 meters with light freezing drizzle and fog, sky totally obscured with vertical visibility 100 feet. There is also moderate icing (clear) in precipitation from surface to 1000 feet AGL and moderate icing in cloud (rime) from 1000 feet AGL up to 10000 feet AGL. The forecast maximum temperature is 00°C at 012100Z and the forecast minimum temperature is minus 01°C at 021200Z.

1.3.8.2. Example Corrected (COR) TAF. [Figure 1.4](#) provides an example of a corrected (COR) TAF for Ramstein AB, Germany. Interpretation and explanation of the coded information follows.

Figure 1.4. Corrected TAF Example.

**TAF COR ETAR 011615Z 0116/0222 28012G25KT 8000 -RASN SCT006 BKN015
OVC020 620158 540009 QNH2960INS
BECMG 0118/0119 27012KT 9999 NSW SCT015 BKN020 QNH2965INS TX15/0120Z
TN04/0211Z**

1.3.8.2.1. The forecast is a correction for Ramstein AB, Germany (ETAR), issued on the first of the month at 1615Z, valid from 011600Z to 022200Z. Initial conditions (011600Z to 011900Z) for the forecast are winds from 280 degrees at 12 knots gusting to 25 knots, visibility 8000 meters in light rain and snow, sky cover is scattered at 600 feet, broken at 1500 feet, and overcast at 2000 feet. There is light rime icing in cloud between 1500 and 9500 feet AGL and occasional moderate turbulence in cloud from surface to 9000 feet AGL. Lowest altimeter setting from 011600Z to 011900Z is 29.60 inches of mercury.

1.3.8.2.2. Between 011800Z and 011900Z, the predominant condition changes gradually to winds from 270 degrees at 12 knots, visibility greater than or equal to 9,999 meters, no significant weather, sky cover scattered at 1500 feet and broken at 2000 feet. The lowest altimeter setting from 011900Z to 022200Z is 29.65 inches of mercury. The forecast maximum temperature is 15°C at 012000Z and the forecast minimum temperature is forecast 4°C at 021100Z.

1.3.9. Operationally significant/Hazardous weather Groups. Volcanic ash and wind shear are potentially hazardous problems for aircraft. Include forecasts for ash and non-convective wind shear on an as-needed basis to focus the attention of the pilot on existing or expected problems.

1.3.9.1. VA. The volcanic ash group indicator. Volcanic Ash (VA) Group (VAbbbbttt). Include a VA group in the TAF, following the cloud and obscuration group. Encode all VA plume forecasts provided by the VAAC in TAF coded products. The VA plume forecast must be horizontally consistent with the official VAAC forecast. **(T-0)** Note: If the responsible VAAC cannot produce the volcanic ash products, then Air Force forecasts are the primary source. **(T-0)**

1.3.9.1.1. bbb. The height of the base of the volcanic ash, encoded in hundreds of feet AGL

1.3.9.1.2. ttt. The height of the top of the volcanic ash layer, encoded in hundreds of feet AGL, as forecast by the VAAC.

1.3.9.1.3. When forecasting VA to be surface based, encode VA as both present weather (w'w') and add a VA group. The following examples show the use of the VA group:

Figure 1.5. Example of Surface-based Volcanic Ash Forecast.

TAF CCCC 101555Z 1016/1122 24010KT 9999 VA FEW100 VA000200 QNH2992INS
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1.3.9.1.4. In this example VA is surface-based and aloft. The VAAC ash plume forecast from the surface with a plume top of 20,000 feet.

Figure 1.6. Example of Volcanic Ash Plume Forecast.

TAF CCCC 101555Z 1016/1122 24010KT 9999 FEW100 VA100200 QNH2992INS

1.3.9.1.5. VA is not surface-based but forecasted in a VAAC ash plume over the TAF location. The VAAC forecasted ash plume has a base height of 10,000 feet and a plume top of 20,000 feet.

1.3.9.2. Non-Convective Low-Level Wind Shear Group (WSHxhxx/dddfffKT). Use this group only to forecast wind shear not associated with convective activity from the surface up to and including 2,000 feet AGL.

1.3.9.2.1. Encode non-convective low-level wind shear forecasts in the following format:

1.3.9.2.1.1. WS. Low-level wind shear group indicator.

1.3.9.2.1.2. hxxhxx. Forecast height of the wind shear in hundreds of feet AGL.

1.3.9.2.1.3. ddd. Forecast wind direction, in tens of degrees true, above the indicated height. Do not use VRB in the non-convective low-level wind shear forecast group.

1.3.9.2.1.4. ff. Forecast wind speed, in knots, of the forecast wind above the indicated height.

1.3.9.2.1.5. KT. Unit indicator for wind speed in knots.

1.3.9.2.2. Non-convective low-level wind shear forecasts are included in the TAF, when expected, following the cloud forecast and before the altimeter setting forecast in the initial forecast period or in a FM or BECMG group. Once included in the forecast, the wind shear group remains the prevailing condition until the next FM or BECMG group or until the end of the forecast valid period if there are no subsequent FM or BECMG groups. Forecasts for non-convective low-level wind shear will not be included in TEMPO groups. **(T-1)**

1.3.9.2.2.1. The following is an example of a TAF containing a non-convective low-level wind shear forecast.

Figure 1.7. Example of TAF with Non-Convective Low-Level Wind Shear.

**TAF CCCC 011555Z 0116/0222 03008KT 0800 PRFG FEW000 BKN005 BKN012
WS015/12038KT QNH3001INS FG FEW000
TEMPO 0118/0120 14012G18KT 3200 -SN BLSN FEW000 OVC006 620065 SN FEW000
FM012130 15012G20KT 9999 NSW SCT030 QNH2992INS
BECMG 0123/0124 15012G20KT 3200 -SN BLSN FEW000 OVC004 620046
QNH2983INS SN FEW000 TX08/0119Z TNM04/0211Z**

1.3.9.2.2.2. In this TAF, non-convective low-level wind shear is forecasted at 1,500 feet with winds from 120 degrees at 38 knots from 011600Z until the beginning of the next FM group at 012130Z.

1.3.10. Icing Group (6IchihihitL). Forecast icing group used to forecast icing not associated with thunderstorms (thunderstorm forecasts imply moderate or greater icing). Repeat as necessary to indicate multiple icing layers. Omit when no icing is forecast. Format icing groups as:

1.3.10.1. 6—Icing group indicator.

1.3.10.2. Ic—Type of icing ([Table 1.5](#)). When forecasting more than one type of icing within the same layer, encode the highest code figure

1.3.10.3. hihhi—Height of base of forecasted icing layer in hundreds of feet AGL ([Table 1.4](#)).

1.3.10.4. tL—Icing layer thickness in thousands of feet ([Table 1.6](#)). When forecasting a layer to be thicker than 9,000 feet, repeat the icing group so that the base of the layer expressed by the second group coincides with the top layer given by the first group (**See Note**).

1.3.11. Turbulence group (5BhBhBhBtL). Forecast turbulence group used only to forecast turbulence not associated with a thunderstorm (thunderstorms already imply severe or extreme turbulence). Turbulence forecasts apply to category II (CAT II) aircraft. A list of aircraft turbulence categories can be found in Air Force Handbook (AFH), 11-203, Volume 2, *Weather for Aircrews – Products and Services*. Omit when no turbulence is forecasted. Format turbulence groups as:

1.3.11.1. 5—Turbulence group indicator.

1.3.11.2. B—Type and intensity of turbulence ([Table 1.7](#))—When forecasting more than one type of turbulence within the same layer, encode the highest code figure

1.3.11.3. hBhBhB—Height of base of forecasted turbulence layer in hundreds of feet AGL ([Table 1.4](#)).

1.3.11.4. tL—Thickness of the turbulence layer in thousands of feet ([Table 1.6](#))—When forecasting a layer to be thicker than 9,000 feet, repeat the turbulence group so that the base of the layer expressed by the second group coincides with the top layer given by the first group. **NOTE:** Icing and turbulence forecasts are for phenomena not associated with thunderstorm activity, from surface to 10,000 feet AGL. Forecasters may address the areas above 10,000 feet mean sea level (MSL) provided the forecast in the TAF is horizontally consistent with turbulence products in the forecaster-in-the-loop (FITL) graphics. Deviations from the authoritative forecast require concurrence from the servicing OWS.

Table 1.4. Height of Lowest Level of Turbulence (hBhBhB)/Icing (hihihi).

Code Figure	Meters	Feet
000	<30	<100
001	30	100
002	60	200
003	90	300
004	120	400
005	150	500
006	180	600
007	210	700
008	240	800
009	270	900
010	300	1,000
011	330	1,100
099	2,970	9,900
100	3,000	10,000
110	3,300	11,000
120	3,600	12,000

Table 1.5. Icing Type (Ic).

Code Figure	Type of Icing
0	Trace icing
1	Light icing (mixed)
2	Light icing in cloud (rime)
3	Light icing in precipitation (clear)
4	Moderate icing (mixed)
5	Moderate icing in cloud (rime)
6	Moderate icing in precipitation (clear)
7	Severe icing (mixed)
8	Severe icing in cloud (rime)
9	Severe icing in precipitation (clear)

Table 1.6. Thickness of Turbulence/Icing Layers (tL).

Code Figure	Thickness
1	1,000 feet
2	2,000 feet
3	3,000 feet
4	4,000 feet
5	5,000 feet
6	6,000 feet
7	7,000 feet
8	8,000 feet
9	9,000 feet

Table 1.7. Turbulence Type/Intensity (B).

Code Figure	Turbulence Type and Intensity
0	None
1	Light Turbulence
2	Moderate Turbulence in clear air, occasional.
3	Moderate Turbulence in clear air, frequent.
4	Moderate Turbulence in cloud, occasional.
5	Moderate Turbulence in cloud, frequent.
6	Severe Turbulence in clear air, occasional.
7	Severe Turbulence in clear air, frequent.
8	Severe Turbulence in cloud, occasional.
9	Severe Turbulence in cloud, frequent.
X	Extreme Turbulence
Note: Occasional is defined to occur less than 1/3 of the time. Frequent is defined as occurring greater than or equal to 1/3 of the time.	

1.3.12. Lowest Altimeter group (QNHP1P1P1INS). Lowest altimeter setting expected (in inches of mercury) during the initial forecast period and in each Becoming (BECMG) and From (FM) change group. Do not include QNH in Temporary (TEMPO) groups. Format the altimeter group as:

1.3.12.1. QNH—Altimeter setting in inches of mercury indicator

1.3.12.2. P1P1P1P1—Forecast lowest altimeter setting

1.3.12.3. INS—Indicator for units of measure for inches of mercury.

1.3.13. TAF Remarks. For weather and obscurations, use the alphabetic abbreviations in [Table 1.2](#) Use Federal Aviation Administration (FAA) Order JO 7340.2, *Contractions*, for all others. Relate operationally significant forecast elements to geographical features within the aerodrome radius when possible and add start/end times without adding a Z when applicable (e.g. FG OVR RIVER E, WND 06010KT AFT 1219, or SHRA OMTNS E 1414-1419). Ensure start/end times are not confused with other numerical values. Do not use the terms OCNL, VC, or CB in remarks. Do not use the remarks section as a substitute for a BECMG or TEMPO group. Encode remarks in the following order of entry:

1.3.13.1. Forecast Maximum and Minimum Temperature groups (T(X)(N)[M]TFTF/YYGFGFZ). This group provides a mechanism to forecast a two-digit temperature (TFTF: whole degrees Celsius) in the TAF code for specific times. To indicate forecast maximum and minimum temperatures expected to occur at the time indicated by GFGFZ, the letter indicator TX for the maximum forecast temperature and TN for the minimum forecast temperature shall precede TFTF without a space. Organizations encode the forecast maximum (first entry) and minimum temperature (last entry) for the first 24-hour period of the TAF. Format temperature groups as:

1.3.13.1.1. TX—Maximum Temperature remark indicator

1.3.13.1.2. TN—Minimum Temperature remark indicator

1.3.13.1.3. TFTF—The forecast temperature in whole degrees Celsius (C). Precede temperatures between +9°C and -9°C with a zero (0); precede temperatures below 0°C by the letter M (for minus).

1.3.13.1.4. YY—The 2 digit day of the month.

1.3.13.1.5. GFGF—The valid time to the nearest whole hour UTC of the temperature forecast.

1.3.13.1.6. Z—Abbreviation for Zulu, the military time zone associated with UTC.

Figure 1.8. Example of Min/Max Temperature Groups.

TX17/0721Z TN08/0812Z — forecast maximum temperature is 17°C at 072100Z and forecast minimum temperature is 8°C at 081200Z.

TX00/1418Z TNM09/1507Z — forecast maximum temperature is 0°C at 141800Z and forecast minimum temperature is minus 9°C at 150700Z.

1.3.13.2. Limited-Duty and Limited METWATCH Remarks. Limited-duty locations coordinate with their supporting Operational Weather Squadron on all TAF valid times around operational hours. At locations where limited-duty operations (i.e. when there is not 24-hour coverage with weather personnel on duty) are in effect, use the following remarks with YY being the day of the month UTC and GG is the time to the nearest whole hour UTC.

1.3.13.2.1. LAST NO AMDS AFT YYGG NEXT YYGG—Use this remark when the airfield is closed and a TAF is no longer required per coordinated requirements.

1.3.13.2.2. LIMITED METWATCH YYGG TIL YYGG—Use this remark when an airfield is open, no weather personnel are on duty and an operational automated sensor is not in use.

Attachment 1**GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION*****References***

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AFH 11-203 V2, *Weather For Aircrews – Products and Services*, 21 May 2018

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AFTTP(I) 3-2.56, Change 1, *Multiservice Tactics, Techniques, and Procedures for Chemical Biological, Radiological, and Nuclear Contamination Avoidance*, 02 February 2006

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Federal Meteorological Handbook No. 12 (FCM-H12), *United States Meteorological Codes and Coding Practices*, 1 January 2013

NATO AWP-4(B), *NATO Meteorological Codes Manual*

World Meteorological Organization (WMO) *Manual on Codes*, No. 306, Volume 1.I, Part A

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

Minus (-)—Light Intensity

Plus (+)—Heavy Intensity

ABV—Above

ADF—Active Dark Filament

AFMAN—Air Force Manual

AFS—Arch Filament System

AGL—Above Ground Level

AKNOW—Event Acknowledgement Code

APR—Active Prominence Region

ASR—Active Surge Region

BC—Patches
BKN—Broken
BL—Blowing
BLO—Below
BR—Mist
BSD—Bright Surge on Disk
BURST—Discrete Solar Radio Burst
BXOUT—Videometer Box Dimension Outline
CA—Cloud to air
CAT—Category
CB—Cumulonimbus
CBRN—Chemical, Biological, Radiological, Nuclear
CC—Cirrocumulus, or Cloud to cloud lightning
CG—Cloud to ground
CHOP—Turbulence type characterized by rapid, rhythmic jolts
CLR—Clear
CONS—Continuous
CONTRAILS—Condensation trails
COR—Correction To A Previously Disseminated Report
CS—Cirrostratus
CU—Characterization Unit
DALAS—Solar Disk and Limb Activity Summary Code
dB—Decibel
DoD—Department Of Defense
DR—Low Drifting
DS—Dust-storm
DSD—Dark Surge on Disk
DSF—Disappearance of a Solar Filament
DSN—Defense Switched Network
DU—Widespread Dust
DURC—During Climb
DURD—During Descent

DZ—Drizzle

EDM—Effective Downwind Message

EPL—Eruptive Prominence on Limb

EST—Estimate, Estimated

EVENT—Event Code

EXTRM—Extreme

FAA—Federal Aviation Administration

FALSE—False Alarm

FC—Funnel Cloud

+FC—Tornado or Waterspout

FEW—Few

FG—Fog

FITL—Forecaster in the Loop

FLARE—Solar Flare Code

FLIP—Flight Information Publication

Fmin—Minimum Observed Frequency

foEs—Sporadic E Critical Frequency

fo F1—F1 Region Critical Frequency

fo F2—F2 Region Critical Frequency

FRQ—Frequent

FU—Smoke

FZ—Freezing

G—Gust

GHz—Giga Hertz (10⁹ Hz)

GMT—Greenwich Mean Time

GPS/NAVSTAR—Global Positioning System/Navigation, Surveillance, Tracking, and Reporting

GR—Hail

GS—Snow Pellets

HLSTO—Hailstone(s)

HSTRY—Histogram History Code

HVY—Heavy

Hz—Hertz

HZ—Haze

IC—Ice Crystals, In-Cloud Lightning

IFLUX—Integrated Solar Radio Flux Code

IMC—Instrument Meteorological Conditions

IMS—Ionospheric Measuring System

INTMT—Intermittent

IONHT—Ionospheric Height Code

IONOS—Automated Ionospheric Data Code

IPP—Ionospheric Penetration Point

KT—Knots

LAST—Last Forecast Before A Break In Coverage At A Manual Station

LGT—Light

LLWS—Low level wind shear

LN—Line

LPS—Loop Prominence

LTG—Lightning

LYR—Layer

M—Meters, Sub-zero temperatures

MANOP—Manual Operations

METAR—Aviation Routine Weather Report

MHz—Mega Hertz (10⁶ Hz)

MI—Shallow

MOD—Moderate

MOV—Moved/Moving/Movement

MSL—Mean Sea Level

OMTNS—Over Mountains

MXD—Mixed

NATO—North Atlantic Treaty Organization

NAVAID—Navigational aids

NE—Northeast

NEG—Negative

NMRS—Numerous

NSW—No Significant Weather
NW—Northwest
OCNL—Occasional
OVC—Overcast
PIREP—Pilot Weather Report
PL—Ice Pellets
PLAIN—Plain Language Code
PO—Dust/Sand Whirls
PY—Spray
RA—Rain
RCA—Reach Cruising Altitude
RFI—Radio Frequency Interference
RSTN—Radio Solar Telescope Network
S4—Mean Amplitude Scintillation Index
SA—Sand
SCT—Scattered
Sec—Second
SEV—Severe
SFC—Surface
Sfu—Solar Flux Units
SG—Snow Grains
SH—Shower(s)
Sigma—sub-delta-phi —Mean Phase Scintillation Index
SKC—Sky Clear
SLD—Solid
SM—Statute Miles
SN—Snow
SOON—Solar Observing Optical Network
SPOTS—Sunspot Code
SPY—Spray
SQ—Squall
SRBL—Solar Radio Burst Locator

SRS—Solar Radio Spectrograph

SS—Sand storm

STATS—Patrol Status Code

STN—Station

SWEEP—Spectral Solar Radio Burst Code

SWPC—Space Weather Prediction Center

TAF—Terminal Aerodrome Forecast

TARWI—Target Weather Information Reporting Code

TBD—To Be Determined

TCU—Towering Cumulus

TEC—Total Electron Current

TEI—Text Element Indicator

TELSI—Total Electron Content and Scintillation Code

TOC—Top of Climb

TOP—Top of Clouds

TS—Thunderstorm

UA—TEI used in routine PIREP

UNKN—Unknown PIREP TEI

UP—Unknown Precipitation

US—United States

USAF—United States Air Force

UTC—Coordinated Universal Time

UUA—TEI used in urgent PIREP

V—Variable

VA—Volcanic Ash

VAAC—Volcanic Ash Advisory Center

VC—Vicinity

VHF—Very High Frequency

VIS—Visibility

VOR—Very high frequency omnidirectional range station

VORTAC—Very high frequency omnidirectional range station/tactical air navigation

VRB—Variable

VV—Vertical Visibility

WMO—World Meteorological Organization

WND—Wind

XR—Solar X-ray Event Indicator

Terms

Bulletin Heading—A combination of letters and numbers that describe the contents of a bulletin, including the data type, geographical location, four-letter identifier of the originator and a date-time group. (e.g. MANOP heading)

Contrails (Condensation trails)—A visible cloud streak, usually brilliant white in color, which trails behind an aircraft or other vehicle in flight under certain conditions.

File Time—The time a weather message or bulletin is scheduled to be transmitted. Expressed either as a specific time or as a specific time block during which the message is transmitted.

Four-letter identifier—A specifically authorized four-letter code assigned to a location and documented by International Civil Aviation Organization.

Issue Time—Time the last agency was notified. Exclude follow-up notifications when determining issue time.

Limited Duty Station—A weather station that provides less than 24-hour a day forecast service.

NAVAID—An electronic navigation aid facility, specifically limited to VHF Omni-Directional Radio Range (VOR), or combined VHF Omni-Directional Radio Range/Tactical Air Navigation (VORTAC) facilities.

Pilot Report—A report of in-flight weather provided by an aircraft crewmember.

Text Element Indicator (TEI)—A two-letter contraction with slash used in the standard PIREP message to identify the elements being reported.

Vicinity—Used to report present weather phenomena when greater than 5 but less than or equal to 10 statute miles of the station.